

Linear Circuits One (EEL 3004C)

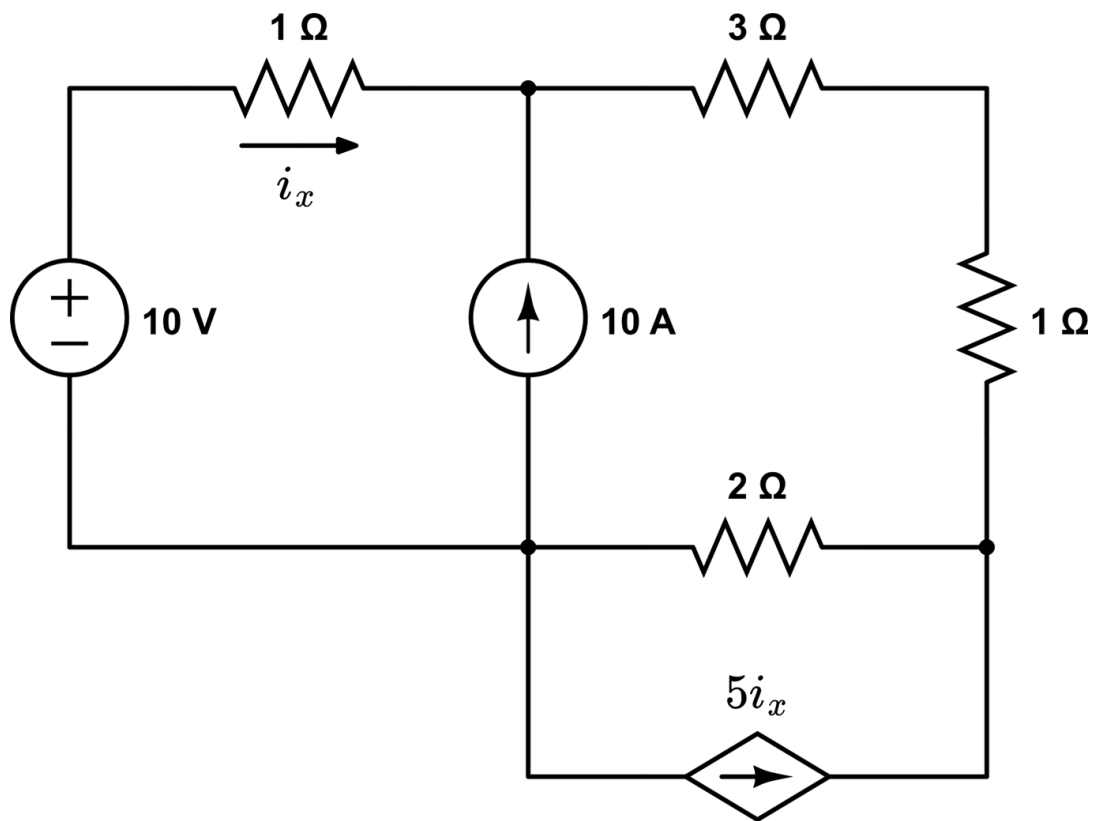
Practice Final Exam

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Question One

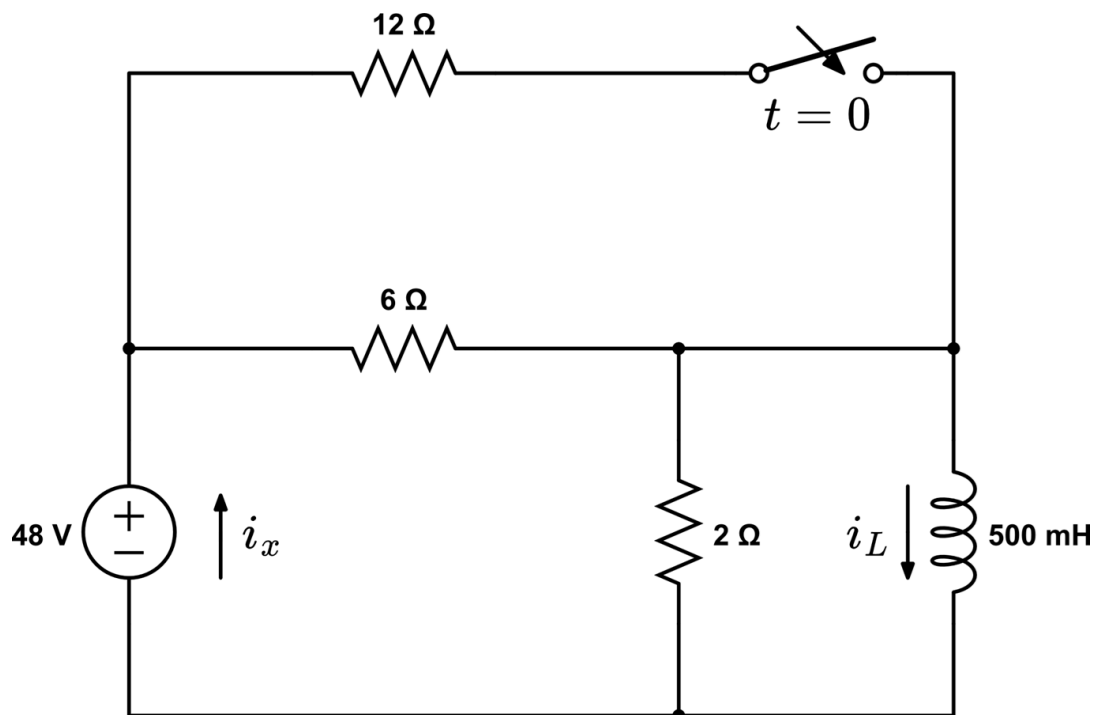
1. Use node-voltage method or mesh-current method to calculate i_x
2. Calculate the power of the dependent source and state whether the power is delivered or absorbed



Question Two

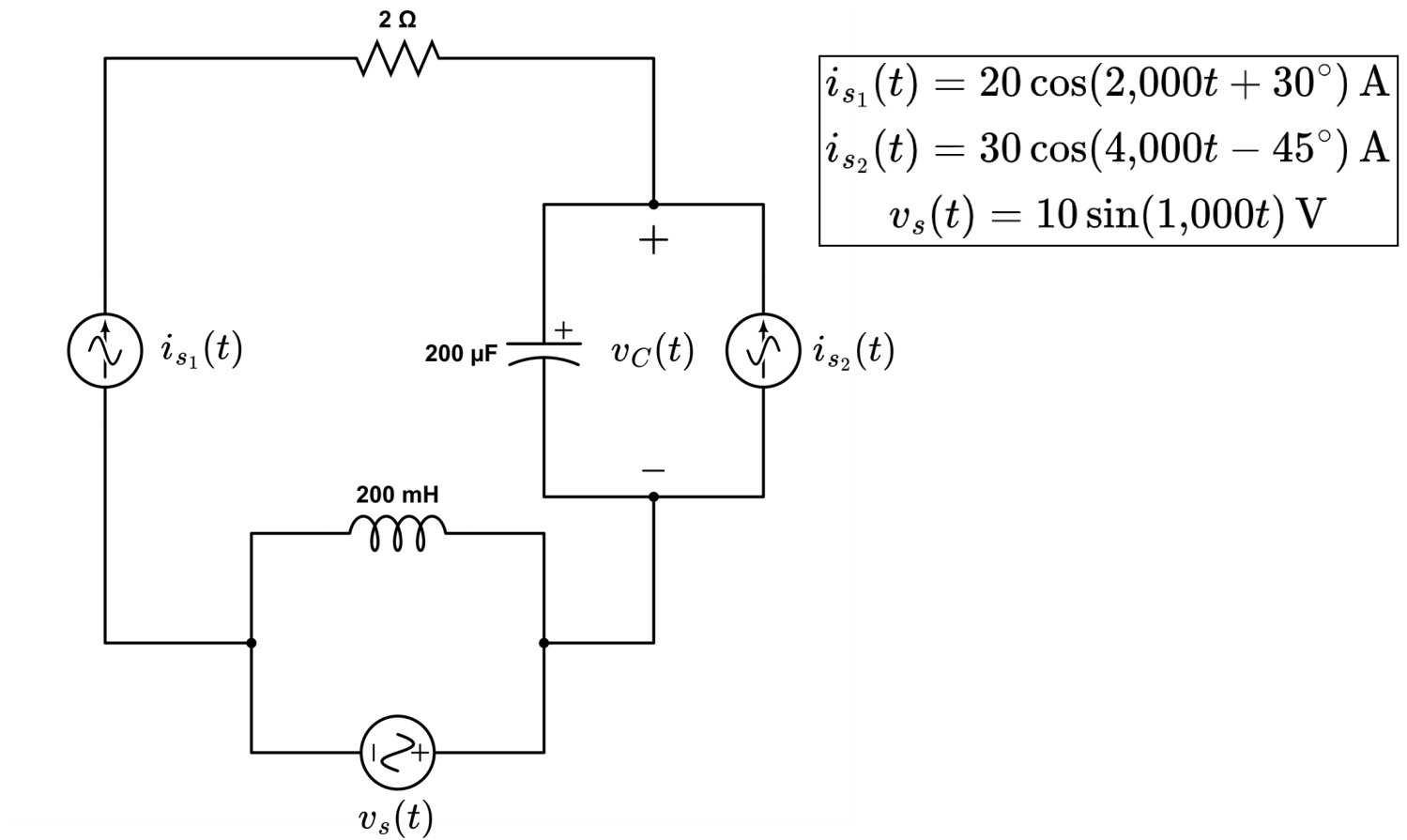
In the circuit below, the switch has been open for a long time and closes at $t = 0$. Determine the following:

1. The initial current in the inductor $i_L(0)$
2. The time domain expression for the inductor current $i_L(t)$, $t \geq 0$
3. The time domain expression for the named current $i_x(t)$, $t \geq 0$



Question Three

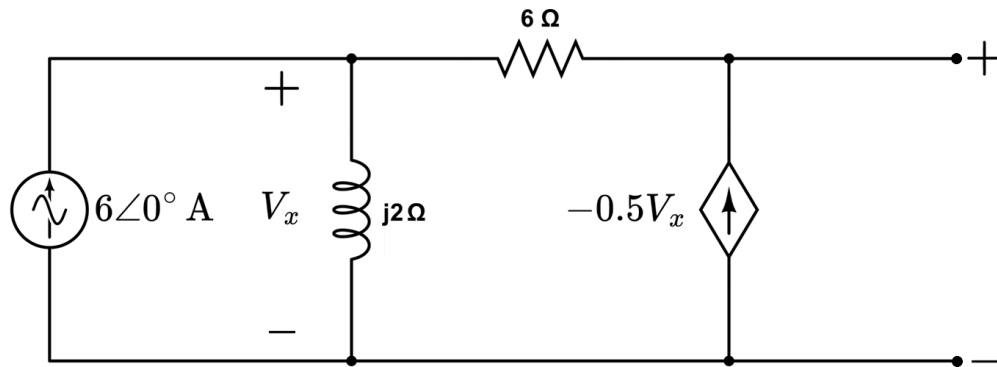
In the circuit below, determine the steady-state expression for the voltage across the capacitor $v_C(t)$



Question Four

For the circuit below, determine the following seen by the open terminal pair:

1. The Thévenin equivalent circuit
2. The impedance that yields maximum average power transfer
3. The impedance that yields maximum average power transfer



Question Five

The circuit below consists of two loads, branch one and branch two. Calculate the following for branch one:

1. The power factor
2. The complex power, represented in rectangular form
3. The average power, state whether the power is delivered or absorbed
4. The reactive power, state whether the power is delivered or absorbed

